

Matthew Maccelari

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RESEARCH PROFILE

Masters of Science in Engineering by dissertation candidate at the University of the Witwatersrand, working in the field of atmospheric optics, applied machine learning and real-time radio systems engineering. My research develops a neural network channel predictor that anticipates turbulence-induced fades on a free-space optical link, enabling pre-emptive link adaptation. My research combines photonics, machine learning and radio systems engineering to support affordable rural connectivity in South Africa. I hold a BSc (Eng) in Electrical Engineering (Information) from Wits, and bring production software engineering experience from a United Kingdom technology firm.

EDUCATION

University of the Witwatersrand Jan 2026 - Dec 2027
MSc (Eng) by Dissertation Johannesburg

- Supervisors: Prof. Mitchell A. Cox and Prof. Ling Cheng, Optical Communication Laboratory.
- Dissertation: *Context-Aware Channel Prediction for Predictive MCS Selection in TVWS over Free-Space Optics*.

University of the Witwatersrand Jan 2022 - Dec 2025
BSc (Eng) Electrical (Information) Engineering Johannesburg

- Placed in 2nd in the final-year class by average mark out of 112 students. Final-year average of 82 percent, fifteen A symbols across the degree, highest mark 94 percent for Software Development.
- Relevant coursework: Machine Learning, Digital Signal Processing, Data Science, Statistics, Control Systems, Software Development, Quantum Computing, Engineering Ethics.
- Elected as School of Electrical and Information engineering student council class representative for final year.

St David's Marist Inanda 2016 - 2021
National Senior Certificate (IEB) Sandton

RESEARCH EXPERIENCE

Optical Communication Laboratory, University of the Witwatersrand Jan 2026 - present
MSc researcher, hybrid free-space optical and TV White Space link adaptation Johannesburg

- **Atmospheric channel modelling:** Built a split-step wave-optics propagation model using von Kármán phase screens with frozen-flow statistics, validated against gamma-gamma turbulence theory in the weak-turbulence regime to within approximately two percent on both the phase structure function and the Rytov scintillation prediction.
- **Machine learning:** Constructed a leakage-controlled training dataset of roughly 33 000 windows and trained a compact recurrent network in PyTorch, currently reducing prediction error by approximately 16 percent against a tuned persistence baseline.
- **Systems integration:** Implementing the model as a real-time channel emulator streaming over ZeroMQ into a GNU Radio software-defined radio flowgraph, working with the laboratory's 800m folded outdoor link and an Ettus USRP B210 running a GFDM transceiver stack.

PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

- M. Maccelari, M. A. Cox and L. Cheng, "A Physics-Validated Digital Twin of a Turbulent Free-Space Optical Link for AI-Driven Adaptation", submitted to the Southern Africa Telecommunication Networks and Applications Conference (SATNAC), 2026.

PROFESSIONAL EXPERIENCE

Isode Ltd Nov 2025 - Jan 2026
Software Engineer Hampton, United Kingdom

- Developed synchronous serial drivers enabling Linux systems to communicate with high-frequency radio modems, working directly in a live production codebase.
- Built a network proxy service for secure external monitoring and remote updating of serial proxy instances.
- Worked to Git-based review workflows in an established multi-contributor codebase, writing tests alongside each feature.

- Done work for Software Development II, Microprocessors, Fundamentals of Communication and Health Systems Dynamics
- Marked laboratory reports and assessments and provided feedback on student performance.
- Organised and led laboratory sessions and tutorials to support student engagement with course material.
- Prepared and delivered guest lectures on selected course topics and introduced students to postgraduate study.

- Wrote Python pipelines to extract, clean and analyse operational log data from onsite machinery, surfacing patterns that helped engineers diagnose recurring production faults.
- Updated microcontroller firmware on the control board of a DockLock system.

SCIENCE OUTREACH AND COMMUNITY ENGAGEMENT

Member of the Optica SPIE Wits Student Chapter: Helped to organise public events and quizzes for the chapter as well as taking part in exhibitions showing off the world of structured light and optical communication systems to schoolchildren. I took part in this at Wit's Yebo Gogga and Johannesburg City Parks and Zoo National Science Month Celebration in Soweto.

Part of the Optical Communications Laboratory: Represented the lab at the National Science Month Launch day at Vaal University of technology as well as gave tours of our lab to industry members and school children.

SELECTED PROJECTS

Bidvest Facilities Management Digital Twin Electronics Lab (*Python, MQTT, TypeScript*)

Industry work alongside Bidvest Facilities Management under the supervision of Professor Ling Cheng where I developed a digital twin of Wit's electronics lab with live metric tracking and LLM integration for smart monitoring of the lab to improve teaching for the students.

PureHerb, live e-commerce platform (*React, Node.js, PostgreSQL*)

Role-based retail platform with separate administrator, distributor and customer interfaces, secure hashed authentication, and the client's full product range. Live and serving users at www.pureherb.co.za.

Quantum Support Vector Machine versus Classical SVM (*Python, Qiskit, IBM Quantum*)

End-to-end binary classification pipeline using PCA reduction, a ZZFeatureMap quantum kernel and QSVC training, benchmarked against a classical RBF-kernel SVM on both simulators and IBM Quantum hardware. Matched classical accuracy of approximately 90 percent on larger samples. Received a project mark of 100 percent.

Free-Space Optical Communication Demonstrator (*C++, 3D CAD Modelling, Electric Circuits*)

Designed and built a small free-space optical communication demonstrator using infrared and visible red light, capable of transmitting data at 100 kb/s. It has been used to demonstrate to schoolchildren how fibre-optic and free-space optical communication systems work in everyday life. Designed and 3D-printed an enclosure to protect the components and encourage curiosity about engineering design. Project writeup available at: <https://www.github.com>

NOTABLE ACHIEVEMENTS

- **Finalist, IEEE Computer Society Global Student Challenge 2026:** Selected among the top five teams internationally from Phase I and invited to present in person to the judging panel at the IEEE Annual Computer Security Applications Conference (ACSAC), Los Angeles, December 2026, with travel support provided by the IEEE Computer Society. Competition post URL: <https://www.computer.org>
- **Third place, CHPC NICIS National Cyber Security Challenge 2024:** Invited to the Gqeberha Boardwalk Hotel and Casino to compete against the top teams in Southern Africa. Also placed 1st for the MATLAB Security Classification Challenge. Competition post URL: www.csc.ac.za
- **University Entrance Scholarship & Postgraduate Merit Award:** University of the Witwatersrand. Earned from National Senior Certificate results (2021) and undergraduate final-year average (2025) respectively.
- **Protea Colours:** Represented South Africa at the 2021 Junior and Under-23 Kayak Sprint World Championships in Montemor-o-Velho, Portugal. Earning national colours for kayaking.

REFEREES

Prof. Mitchell A. Cox (supervisor) • School of Electrical and Information Engineering, Wits • mitchell.cox@wits.ac.za
Prof. Ling Cheng (co-supervisor) • School of Electrical and Information Engineering, Wits • ling.cheng@wits.ac.za